

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Selects appropriate method for example by changing to reduced equation by dividing by $\tan x$	AO3.1a	M1	$\frac{dy}{dx} + (\cot x)y = \sin x$
	Finds correct integrating factor	AO1.1b	B1	Integrating factor $= e^{\int (\cot x) dx}$ $= e^{\ln(\sin x)} = \sin x$
ALT	Alt. finds an integrating factor by inspection, using original equation. (PI)	AO3.1a	M1	$\cos x \tan x \frac{dy}{dx} + y \cos x = \sin x \tan x \cos x$ $\sin x \frac{dy}{dx} + (\cos x)y = \sin^2 x$
	Finds integrating factor = $\cos x$	AO1.1b	B1	
	Multiplies reduced or original equation by 'their' integrating factor and identifies LHS as differential of $y \times \sin x$ PI	AO1.1a	M1	$\sin x \frac{dy}{dx} + (\cos x)y = \sin^2 x$ $\frac{d}{dx} [y \sin x] = \sin^2 x$
	Uses appropriate integration method for RHS of 'their' equation	AO1.1a	M1	$y \sin x = \int \left\{ \frac{1}{2} (1 - \cos 2x) \right\} dx$
	Integrates correctly to obtain correct solution	AO1.1b	A1	$y \sin x = \frac{1}{2} x - \frac{1}{4} \sin 2x + C$
	Uses boundary condition after integration completed in either $y \sin x = \dots$ or $y = \dots$ form OE	AO1.1a	M1	$\sin \frac{\pi}{4} \cdot \frac{1}{2\sqrt{2}} = \frac{1}{2} \cdot \frac{\pi}{4} - \frac{1}{4} \sin \frac{\pi}{2} + C$ $C = \frac{1}{2} - \frac{\pi}{8}$
(b)	States fully correct particular solution	AO1.1b	A1	$y \sin x = \frac{1}{2} x - \frac{1}{4} \sin 2x + \frac{1}{2} - \frac{\pi}{8}$
Total 7 marks				

Q2.

(a) $\frac{dy}{dx} + \frac{\sec^2 x}{\tan x} y = \tan x$

IF is $\exp\left(\int \frac{\sec^2 x}{\tan x} dx\right)$
and with integration attempted

M1

$= e^{\ln(\tan x)} = \tan x$
AG Be convinced

A1

2

(b) $\tan x \frac{dy}{dx} + (\sec^2 x)y = \tan^2 x$

$\frac{d}{dx} [y \tan x] = \tan^2 x$
LHS as differential of $y \times IF$ PI

M1

$y \tan x = \int \tan^2 x dx$

A1

$\Rightarrow y \tan x = \int (\sec^2 x - 1) dx$
Using $\tan^2 x = \sec^2 x - 1$ PI or other valid methods to integrate $\tan^2 x$

m1

$y \tan x = \tan x - x (+c)$
Correct integration of $\tan^2 x$; condone absence of $+c$.

A1

$3 \tan \frac{\pi}{4} = \tan \frac{\pi}{4} - \frac{\pi}{4} + c$

Boundary condition used in attempt to find value of c

m1

$c = 2 + \frac{\pi}{4}$ so $y \tan x = \tan x - x + 2 + \frac{\pi}{4}$

$y = 1 + (2 - x + \frac{\pi}{4}) \cot x$
ACF

A1

Q3.

(a) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = 3x - 8e^{-3x}$

P. Integral: $y \text{ PI} = a + bx + cxe^{-3x}$

$y' \text{ PI} = b + ce^{-3x} - 3cxe^{-3x}$

*Product rule used at least once giving terms in the form
 $\pm pe^{-3x} \pm qxe^{-3x}$*

M1

$y'' \text{ PI} = -6ce^{-3x} + 9cxe^{-3x}$

$-6ce^{-3x} + 9cxe^{-3x} + 2b + 2ce^{-3x} - 6cxe^{-3x}$

$-3a - 3bx - 3cxe^{-3x} = 3x - 8e^{-3x}$

Substitution into LHS of DE

M1

$-3b = 3; 2b - 3a = 0; -4c = -8$

*Dep on 2nd M only Equating coeffs to obtain at least two
of these correct eqns; PI by correct values for at least two
constants*

m1

$b = -1; c = 2; a = -\frac{2}{3}$

*Dep on M1M1m1 all awarded A1 if any two correct;
A2 if all three correct but do not award the 2nd A mark
if terms in xe^{-3x} were incorrect in the M1 line*

A2,1,0

$[y \text{ PI} = -\frac{2}{3} - x + 2xe^{-3x}]$

5

(b) Aux. eqn. $m^2 + 2m - 3 = 0$

$(m + 3)(m - 1) = 0$

*Factorising or using quadratic formula OE PI by correct
two values of 'm' seen / used*

M1

$(y_{CF} =) Ae^{-3x} + Be^x$

A1

$$(y_{GS} =) Ae^{-3x} + Be^x - \frac{2}{3} - x + 2xe^{-3x}$$

c's CF + c's PI with 2 arbitrary constants, non-zero values for a, b and c and no trig or ln terms in c's CF

B1F

3

(c) $x = 0, y = 1 \Rightarrow 1 = A + B - \frac{2}{3}$

Only ft if previous B1F has been awarded

B1F

$$\frac{dy}{dx} = -3Ae^{-3x} + Be^x - 1 + 2e^{-3x} - 6xe^{-3x}$$

As $x \rightarrow \infty$, ($e^{-3x} \rightarrow 0$ and) $xe^{-3x} \rightarrow 0$

Must treat xe^{-3x} separately

E1

(As $x \rightarrow \infty$, $\frac{dy}{dx} \rightarrow -1$ so) $B = 0$

B = 0, where B is the coefficient of e^x .

B1

When $B = 0$, $1 = A - \frac{2}{3} \Rightarrow A = \frac{5}{3}$

$$y = \frac{5}{3} e^{-3x} - \frac{2}{3} - x + 2xe^{-3x}$$

A1

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4

[12]

Q4.

(a) $\frac{d}{dx} [\ln(\ln x)] = \frac{1}{\ln x} \times \frac{1}{x}$

ACF

B1

1

(b) (i) $\frac{dy}{dx} + \frac{1}{x \ln x} y = 9x^2$

An IF is $\exp \left\{ \int [1 / (x \ln x)] (dx) \right\}$

.... and with integration attempted

M1

$$= e^{\ln(\ln x)} = \ln x$$

AG Must see $e^{\ln(\ln x)}$ before $\ln x$

A1

2

(ii) $\ln x \frac{dy}{dx} + \frac{1}{x} y = 9x^2 \ln x$

$$\frac{d}{dx} [y \ln x] = 9x^2 \ln x$$

LHS as differential of $y \times \ln x$ PI

M1

$$y \ln x = \int 9x^2 \ln x \, dx$$

A1

$$\Rightarrow y \ln x = \int \ln x \, d[3x^3] = 3x^3 \ln x - \int 3x^3 \left(\frac{1}{x}\right) dx$$

$$\int kx^2 \ln x \, (dx) = px^3 \ln x - \int px^3 \left(\frac{1}{x}\right) (dx) \text{ or better}$$

m1

$$y \ln x = 3x^3 \ln x - x^3 (+c)$$

ACF Condone missing '+c'

A1

When $x = e$, $y = 4e^3$, $4e^3 = 3e^3 - e^3 + c$

Dep on previous M1m1. Boundary condition used in attempt to find value of 'c' after integration is completed.

m1

$$c = 2e^3$$

$$\Rightarrow y \ln x = 3x^3 \ln x - x^3 + 2e^3$$

$$y = 3x^3 - \frac{(x^3 - 2e^3)}{\ln x}$$

ACF

A1

6

[9]

Q5.

(a) IF is $\exp\left(\int \frac{2}{x} dx\right)$

M1

$$= e^{2\ln x}$$

PI

A1

$$= x^2$$

A1

$$\frac{d}{dx}[yx^2] = x^2 \ln x$$

LHS; PI

M1

$$\Rightarrow yx^2 = \int (\ln x) \frac{d}{dx} \left(\frac{x^3}{3} \right)$$

Attempt integration by parts in correct direction to integrate $x^p \ln x$

M1

$$= \frac{x^3}{3} \ln x - \int \frac{x^3}{3} dx$$

RHS

A1

$$yx^2 = \frac{x^3}{3} \ln x - \frac{x^3}{9} + A$$

$$\{y = \frac{x}{3} \ln x - \frac{x}{9} + Ax^{-2}\}$$

A1

3

(b) Now, as $x \rightarrow 0$, $x^k \ln x \rightarrow 0$

Must be stated explicitly for a value of $k > 0$

E1

As $x \rightarrow 0$, $y \rightarrow 0 \Rightarrow A = 0$

Const of int = 0 must be convincing

B1

$$yx^2 = \frac{x^3}{3} \ln x - \frac{x^3}{9}$$

When $x = 1$, $y = -\frac{1}{9}$

ft on one slip but must have made a realistic attempt to find A

B1F

3

[10]

Q6.

(a) IF is $e^{\int \frac{4}{2x+1} dx}$
PI

M1

$$e^{2 \ln(2x+1)} = e^{\ln(2x+1)^2}$$

Either O.E. Condone missing '+c'

A1

$$= (A)(2x+1)^2$$

Ft on earlier $e^{\ln(2x+1)}$, condone missing 'A'

A1F

$$(2x+1)^2 \frac{dy}{dx} + 4(2x+1)y = 4(2x+1)^7$$

$$\frac{d}{dx}[(2x+1)^2 y] = 4(2x+1)^7$$

LHS as $d/dx (y \times c's IF)$ PI and also RHS of form $p(2x+1)^q$

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M1

$$(2x+1)^2 y = \int 4(2x+1)^7 dx$$

A1

$$(2x+1)^2 y = \frac{1}{8} (2x+1)^8 + c$$

Correct integration of $p(2x+1)^q$ to

$$\frac{p(2x+1)^{q+1}}{2(q+1)} \quad (+c) \text{ ft for } q > -2 \text{ only}$$

B1F

(GS): $y = \frac{1}{8} (2x+1)^6 + c (2x+1)^{-2}$

Must be in the form $y = f(x)$, where $f(x)$ is ACF

A1

(b) $y = \frac{1}{4} (2x + 1)^6 + c(2x + 1)^{-2}$

When $x = 0$, $\frac{dy}{dx} = 0$

Using boundary condition $x = 0$, $\frac{dy}{dx} = 0$
and c 's GS in (a) towards obtaining a value for c

M1

$$\Rightarrow y = 1 \left[\frac{dy}{dx} 3 (2x + 1)^5 - 4c(2x + 1)^{-3} \right]$$

Either $y = 1$ or correct expression for
 dy / dx in terms of x only

B1

$$\Rightarrow c = \frac{3}{4} \text{ so } y = \frac{1}{4} (2x + 1)^6 + \frac{3}{4} (2x + 1)^{-2}$$

CSO

A1

3

[10]

Q7.

(a) $p \cos x - q \sin x + 5 p \sin x + 5 q \cos x = 13 \cos x$

Differentiation and subst. into DE

M1

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$$p + 5q = 13; \quad 5p - q = 0$$

Equating coeffs.

m1

$$p = \frac{1}{2}; \quad q = \frac{5}{2}$$

OE Need both

A1

3

(b) Aux. eqn. $m + 5 = 0$

PI. Or solving $y'(x) + 5y = 0$ as far as $y =$

M1

$$(y_{CF}) = Ae^{-5x}$$

A1

$$(y_{CF}) Ae^{-5x} + \frac{1}{2} \sin x + \frac{5}{2} \cos x$$

c's CF + c's PI with exactly one arbitrary constant OE

B1F

3

[6]

Q8.

(a) PI: $y_{PI} = p + qxe^{-2x}$

$$y'_{PI} = qe^{-2x} - 2qxe^{-2x}$$

Product rule used

M1

$$y''_{PI} = -4qe^{-2x} + 4qxe^{-2x}$$

$$-4q e^{-2x} + 4qxe^{-2x} + qe^{-2x} - 2qxe^{-2x}$$

$$-2p - 2qxe^{-2x} = 4 - 9e^{-2x}$$

Subst. into DE

M1

$$-3q = -9 \text{ and } -2p = 4$$

Equating coefficients

©2025 Exam Papers Practice. All rights reserved. m1.

$$-3q = -9 \text{ so } q = 3 ;$$

A1

$$-2p = 4 \text{ so } p = -2 ;$$

$$[y_{PI} = 3x e^{-2x} - 2]$$

B1

5

(b) Aux. eqn. $m^2 + m - 2 = 0$

$$(m - 1)(m + 2) = 0$$

Factorising or using quadratic formula OE

PI by correct two values of 'm' seen/used

M1

$$y_{CF} = Ae^x + Be^{-2x}$$

A1

$$y_{GS} = Ae^x + Be^{-2x} + 3xe^{-2x} - 2$$

$(y_{GS}) = c's\ CF + c's\ PI, \text{ provided } 2\ \text{arbitrary constants}$

B1F

3

(c) $x = 0, y = 4 \Rightarrow 4 = A + B - 2$

Only fit if exponentials in GS

B1F

$$\frac{dy}{dx} = Ae^x - 2Be^{-2x} + 3e^{-2x} - 6xe^{-2x}$$

As $x \rightarrow \infty$, ($e^{-2x} \rightarrow 0$ and) $xe^{-2x} \rightarrow 0$

E1

As $x \rightarrow \infty$, $\frac{dy}{dx} \rightarrow 0$ so $A = 0$

B1

When $A = 0$, $4 = 0 + B - 2 \Rightarrow B = 6$

$$y = 6e^{-2x} + 3xe^{-2x} - 2$$

$$y = 6e^{-2x} + 3xe^{-2x} - 2 \quad \text{OE}$$

B1

4

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[12]

Q9.

$$\frac{dy}{dx} + (\cot x)y = \sin 2x$$

IF is $\exp \left(\int \cot x dx \right)$

and with integration attempted

M1

$$= e^{\ln(\sin x)} (+C)$$

OE Condone missing '+c'

A1

$$= (k) \sin x$$

'IF = sin x' scores M1A1A1

A1

$$\sin \frac{dy}{dx} + (\cos x)y = \sin 2x \sin x$$

$$\frac{d}{dx}[y \sin x] = \sin 2x \sin x$$

LHS as differential of y×IF PI

M1

$$y \sin x = \int \sin 2x \sin x \, dx$$

Ft on c's IF provided no exp. or logs

A1F

$$\Rightarrow y \sin x = \int 2 \sin^2 x \cos x \, dx$$

sin 2x = 2 sin x cos x used

B1

$$\Rightarrow y \sin x = \int 2 \sin^2 x \, d(\sin x)$$

dep on both Ms

*Use of relevant substitution to stage $\int 2s^2 \, ds$
 or further or by inspection to $\int \sin^3 x$*

m1

$$y \sin x = \frac{2}{3} \sin^3 x (+c)$$

ACF

A1

$$\frac{1}{2} \sin \frac{\pi}{6} = \frac{2}{3} \sin^3 \frac{\pi}{6} + c$$

dep on both Ms

*Boundary condition used in attempt to find
 value of c after integration*

m1

$$c = \frac{1}{6} \text{ so } y \sin x = \frac{2}{3} \sin^3 x + \frac{1}{6}$$

CSO – no errors seen – accept equivalent forms

Q10.

(a) $y_{PI} = pxe^{-2x} \Rightarrow \frac{dy}{dx} = pe^{-2x} - 2pxe^{-2x}$

Product Rule used

M1

$$\Rightarrow \frac{d^2y}{dx^2} = -2pe^{-2x} - pe^{-2x} + 4pxe^{-2x}$$

A1

$$-4pe^{-2x} + 4pxe^{-2x} + 3pe^{-2x} - 6pxe^{-2x} + 2pxe^{-2x} = 2e^{-2x}.$$

Sub. into DE

M1

$$-pe^{-2x} = 2e^{-2x} \Rightarrow p = -2$$

ft one slip in differentiation

A1F

4

(b) Aux. eqn. $m^2 + 3m + 2 = 0$
 $\Rightarrow m = -1, -2$

B1

CF is $Ae^{-x} + Be^{-2x}$

ft on real values of m only

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M1

GS $y = Ae^{-x} + Be^{-2x} - 2xe^{-2x}$.

Their CF + their PI must have 2 arb consts

B1F

When $x = 0, y = 2 \Rightarrow A + B = 2$

Must be using GS; ft on wrong non-zero values for p and m

B1F

$$\frac{dy}{dx} = -Ae^{-x} - 2Be^{-2x} - 2e^{-2x} + 4xe^{-2x}$$

Must be using GS; ft on wrong non-zero values for p and m

B1F

When $x = 0$, $\frac{dy}{dx} = 0 \Rightarrow -A - 2B - 2 = 0$
Must be using GS; ft on wrong non-zero values for p and m and slips in finding y'(x)

B1F

Solving simultaneously, 2 eqns each in two arbitrary constants

m1

$A = 6, B = -4; y = 6e^{-x} - 4e^{-2x} - 2xe^{-2x}.$
 CSO

A1

8

[12]

Q11.

Aux. eqn. $m^2 + 4 = 0 \Rightarrow m = \pm 2i$

B1

CF is $A \cos 2x + B \sin 2x$

OE. If m is real give M0

M1

ft on incorrect complex value for m

A1F

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PI: Try $ax^2 + b + c \sin x$

Award even if extra terms, provided the relevant coefficients are shown to be zero.

M1M1

$2a - c \sin x + 4ax^2 + 4b + 4c \sin x = 8x^2 + 9 \sin x$
 $a = 2, b = -1,$

Dep on relevant M mark

A1

$c = 3$

Dep on relevant M mark

A1

$(y =) A \cos 2x + B \sin 2x + 2x^2 - 1 + 3 \sin x$



Their CF + their PI. Must be exactly two arbitrary constants

B1F

[8]

Q12.

(a) $-4k \sin 2x + k \sin 2x = \sin 2x$

Substituting into the differential equation

M1A1

$$k = -\frac{1}{3}$$

Accept correct PI

A1

3

(b) (Aux. eqn $m^2 + 1 = 0$) $m = \pm i$

PI

B1

CF: $A \cos x + B \sin x$

M0 if m is real

M1

OE Ft on incorrect complex values for m

For the A1F do not accept if left in the form $Ae^{ix} + Be^{-ix}$

A1F

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(GS: $y =$) $A \cos x + B \sin x - \frac{1}{3} \sin 2x$

c 's CF + c 's PI but must have 2 constants

B1F

4

[7]

Q13.

IF is $\exp\left(\int \frac{3}{x} dx\right)$

and with integration attempted

M1

$$= e^{3 \ln x}$$

PI

A1

$$= x^3$$

A1

$$\frac{d}{dx}[yx^3] = x^3(x^4 + 3)^{\frac{3}{2}}$$

LHS. Use of c's IF. PI

M1A1

$$\Rightarrow yx^3 = \frac{1}{10}(x^4 + 3)^{\frac{5}{2}} + A$$

$$k(x^4 + 3)^{\frac{5}{2}}$$

m1

Condone missing 'A'

A1

$$\Rightarrow \frac{1}{5} = \frac{1}{10}(4)^{\frac{5}{2}} + A$$

Use of boundary conditions in attempt to find constant after intgr. Dep on two M marks, not dep on m

m1

$$\Rightarrow A = -3; \quad (*)$$

$$\Rightarrow yx^3 = \frac{1}{10}(x^4 + 3)^{\frac{5}{2}} - 3$$

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ACF. The A1 can be awarded at line (*) provided a correct earlier eqn in y, x and 'A' is seen immediately before boundary conditions are substituted.

A1

[9]

Q14.

(a) IF is $e^{\int -\frac{2}{x} dx}$

$$e^{\int \pm \frac{2}{x} dx}$$

M1

$$= e^{-2 \ln x}$$

PI

A1

$$= e^{\ln x^{-2}} = x^{-2} = \frac{1}{x^2}$$

AG Be convinced

A1

3

(b) $\frac{d}{dx} \left(\frac{y}{x^2} \right) = \frac{1}{x^2} x$

LHS as $d/dx (y \times IF)$

M1

PI

A1

$$\frac{y}{x^2} = \int \frac{1}{x} dx = \ln x + c$$

RHS Condone missing '+ c' here

A1

$$y = x^2 \ln x + cx^2$$

A1

4

[7]

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Q15.

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IF is $e^{\int -\tan x dx}$

Award even if negative sign missing

M1

$$= e^{\ln(\cos x) + c}$$

OE Condone missing c

A1

$$= (k) \cos x$$

ft earlier sign error

A1F

$$\cos x \frac{dy}{dx} - y \tan x \cos x = 2 \sin x \cos x$$

$$\frac{d}{dx}(y \cos x) = 2 \sin x \cos x$$

$$\text{LHS as } \frac{d}{dx}(y \times IF) \quad \text{PI}$$

M1

$$y \cos x = \int 2 \sin x \cos x \, dx$$

ft on c's IF provided no exp or logs

A1F

$$y \cos x = \int \sin 2x \, dx$$

Double angle or substitution OE for integrating $2 \sin x \cos x$

m1

$$y \cos x = -\frac{1}{2} \cos 2x (+c)$$

ACF

A1

$$2 = -\frac{1}{2} + c$$

Boundary condition used to find c

m1

$$c = \frac{5}{2}$$

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$$y \cos x = -\frac{1}{2} \cos 2x + \frac{5}{2}$$

*ACF eg $y \cos x - 2 + \sin^2 x$
Apply ISW after ACF*

A1

[9]

Q16.

$$(a) \quad -k \sin x + 2k \cos x + 5k \sin x = 8 \sin x + 4 \cos x$$

Differentiation and subst. into DE

M1A1

$$k = 2$$

A1

(b) Auxl eqn $m^2 + 2m + 5 = 0$

$$m = \frac{-2 \pm \sqrt{4 - 20}}{2}$$

Formula or completing sq. PI

M1

$$m = -1 \pm 2i$$

A1

CF: $\{y_C\} = e^{-x}(A \sin 2x + B \cos 2x)$

ft provided m is not real

A1F

GS $\{y\} = e^{-x}(A \sin 2x + B \cos 2x) + k \sin x$

ft on CF + PI; must have 2 arb consts

B1F

When $x = 0, y = 1 \Rightarrow B = 1$

B1F

$$\frac{dy}{dx} = -e^{-x}(A \sin 2x + B \cos 2x) + e^{-x}(2A \cos 2x - 2B \sin 2x) + k \cos x$$

Product rule

M1

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When $x = 0, \frac{dy}{dx} = 4 \Rightarrow 4 = -B + 2A + k$

PI

A1

$$\Rightarrow A = \frac{3}{2}$$

$$y = e^{-x} \left(\frac{3}{2} \sin 2x + \cos 2x \right) + 2 \sin x$$

CSO

A1

Q17.

(a) $(m + 2)^2 = -1$

Completing sq or formula

M1

$$m = -2 \pm i$$

A1

CF is $e^{-2x} (A \cos x + B \sin x)$

If m is real give M0

M1

{or $e^{-x} A \cos (x + B)$

but not $Ae^{(-1+i)x} + Be^{(-1-i)x}$ }

Ft on wrong a's and b's but roots must be complex

A1ft

PI try $y = p \Rightarrow 5p = 5$ PI is $y = 1$

B1

GS $y = e^{-2x} (A \cos x + B \sin x) + 1$

Their CF + their PI with two arbitrary constants.

B1ft

6

(b) $x = 0, y = 2 \Rightarrow A = 1$

Provided previous B1ft awarded

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B1ft

$$y'(x) = -2e^{-2x} (A \cos x + B \sin x) + e^{-2x} (-A \sin x + B \cos x)$$

Product rule used

M1A1ft

$$y'(0) = 3 \Rightarrow 3 = -2A + B \Rightarrow B = 5$$

Ft on one slip

A1ft

$$y = e^{-2x} (\cos x + 5 \sin x) + 1$$

4

[10]

Q18.

IF is $e^{\int \frac{4x}{x^2+1} dx}$

M1

$$= e^{2 \ln(x^2+1)}$$

A1

$$= e^{\ln(x^2+1)^2} = (x^2 + 1)^2$$

For $e^{2 \ln(x^2+1)}$

A1ft

$$\frac{d}{dx} (y(x^2 + 1)^2) = x(x^2 + 1)^2$$

*LHS as $d/dx (y \times \text{const's IF})$ PI and also
RHS of form $kx(x^2 + 1)^p$*

M1A1ft

$$x(x^2 + 1)^2 = \int x(x^2 + 1)^2 dx$$

$$x(x^2 + 1)^2 = \frac{1}{6} (x^2 + 1)^3 + c$$

*Use of suitable substitution to find RHS
or reaching $k(x^2 + 1)^3$ OE
Condone missing c*

M1A1

$$y(0) = 1 \Rightarrow c = \frac{5}{6}$$

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m1

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$$y = \frac{1}{6} (x^2 + 1) + \frac{5}{6(x^2 + 1)^2}$$

Accept other forms of $f(x)$

$$\text{eg } y = \frac{\left(\frac{x^6}{6} + \frac{2x^4}{4} + \frac{x^2}{2} + 1 \right)}{(x^2 + 1)^2}$$

A1

[9]

Q19.

- (a) PI: $y_{PI} = a + bx + c \sin x + d \cos x$
 $y'_{PI} = b + c \cos x - d \sin x$
 $b + c \cos x - d \sin x - 3a - 3bx - 3c \sin x - 3d \cos x$
 $= 10 \sin x - 3x$

Substituting into DE

M1

$$b - 3a = 0; -3b = -3; c - 3d = 0; -d - 3c = 10$$

Equating coefficients (at least 2 eqns)

M1

$$a = \frac{1}{3}; b = 1; c = -3; d = -1$$

A1 for any two correct

A2,1

$$y_{PI} = \frac{1}{3} + x - 3 \sin x - \cos x$$

4

(b) Aux. eqn. $m - 3 = 0$

Altn. $\int y^{-1} dy = \int 3 dx$ OE (M1)

M1

$$(y_{CF} =) Ae^{3x}$$

$$Ae^{3x} \quad \text{OE}$$

A1

$$(y_{GS} =) Ae^{3x} + \frac{1}{3} + x - 3 \sin x - \cos x$$

(c's CF + c's PI) with 1 arbitrary constant

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B1F

3

[7]

Q20.

(a) $u = \frac{dy}{dx} \Rightarrow \frac{du}{dx} = \frac{d^2y}{dx^2}$

M1

$$x \frac{du}{dx} - u = 3x^2 \Rightarrow \frac{du}{dx} - \frac{1}{x}u = 3x$$

AG Substitution into LHS of DE and completion

A1

2

- (b) IF is $\exp\left(\int -\frac{1}{x} dx\right)$
and with integration attempted

M1

$$= e^{-\ln x}$$

A1

$$= x^{-1} \text{ or } \frac{1}{x}$$

or multiple of x^{-1}

A1

$$\frac{d}{dx}[ux^{-1}] = 3$$

LHS as differential of $u \times \text{IF}$. PI

M1

$$\Rightarrow ux^{-1} = 3x + A$$

Must have an arbitrary constant
(Dep. on previous M1 only)

m1

$$u = 3x^2 + Ax$$

A1

6

(c) $\frac{dy}{dx} = 3x^2 + Ax$

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Replaces u by $\frac{dy}{dx}$ and attempts to integrate

M1

$$y = x^3 + \frac{Ax^2}{2} + B$$

ft on cand's u but solution must have two
arbitrary constants

A1F

2

[10]

Q21.

- (a) Aux eqn: $m^2 - 2m - 3 = 0$

M1

$$m = -1, 3$$

PI

A1

$$CF (y_C =) Ae^{3x} + Be^{-x}$$

M1

$$\text{Try } (y_{PI} =) a e^{-2x} (+ b)$$

M1

$$\frac{dy}{dx} = -2ae^{-2x}$$

A1

$$\frac{d^2y}{dx^2} = 4ae^{-2x}$$

A1

Substitute into DE gives

$$4ae^{-2x} + 4ae^{-2x} - 3ae^{-2x} - 3b = 10e^{-2x} - 9$$

M1

$$\Rightarrow a = 2$$

A1

$$b = 3$$

B1

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$$(y_{GS} =) Ae^{3x} + Be^{-x} + 2e^{-2x} + 3$$

(c's CF+c's PI) with 2 arbitrary constants

B1F

10

$$(b) \quad x = 0, y = 7 \Rightarrow 7 = A + B + 2 + 3$$

Only fit if exponentials in GS and two arbitrary constants remain

B1F

$$\frac{dy}{dx} = 3Ae^{3x} - Be^{-x} - 4e^{-2x}$$

$$\text{As } x \rightarrow \infty, e^{-kx} \rightarrow 0, \frac{dy}{dx} \rightarrow 0 \text{ so } A = 0$$

B1

When $A = 0$, $5 = 0 + B + 3 \Rightarrow B = 2$

Must be using 'A' = 0

B1F

$$y = 2e^{-x} + 2e^{-2x} + 3$$

CSO

A1

4

[14]

Q22.

(a) IF is $\exp\left(\int \frac{2}{x} dx\right)$

And with integration attempted

$$= e^{2\ln x}$$

M1

$$= x^2$$

A1

CSO **AG** be convinced

A1

3

(b) $\frac{d}{dx}[yx^2] = 3x^2(x^3 + 1)^{\frac{1}{2}}$

PI

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M1A1

$$\Rightarrow yx^2 = \frac{2}{3}(x^3 + 1)^{\frac{3}{2}} + A$$

$$k(x^3 + 1)^{\frac{3}{2}}$$

m1

Condone missing 'A'

A1

$$\Rightarrow 4 = \frac{2}{3}(9)^{\frac{3}{2}} + A$$

Use of boundary conditions to find constant

m1

$$\Rightarrow A = -14$$

$$\Rightarrow y = x^{-2} \left\{ \frac{2}{3} (x^3 + 1)^{\frac{3}{2}} - 14 \right\}$$

Any correct form

A1

6

[9]

Q23.

Auxl. eqn $m^2 - 4m + 3 = 0$

PI

M1

$m = 3$ and 1

PI

A1

CF is $A e^{3x} + B e^x$

A1F

PI Try $y = a + b \sin x + c \cos x$

Condone 'a' missing here

M1

$y'(x) = b \cos x - c \sin x$

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A1

$y''(x) = -b \sin x - c \cos x$

ft can be consistent sign error(s)

A1F

Substitute into DE gives

M1

$a = 2$

B1

$4c + 2b = 5$ and $2c - 4b = 0$

A1

$b = 0.5,$

ft a slip

A1F

$$c = 1$$

ft a slip

A1F

$$\text{GS: } y = A e^{3x} + B e^x + 2 + 0.5 \sin x + \cos x$$

*y = candidate's CF and candidate's PI
(must have exactly two arbitrary constants)*

B1F

12

[12]

Q24.

$$(a) \quad y_{PI} = kx^2 e^{5x} \Rightarrow y' = 2kx e^{5x} + 5kx^2 e^{5x}$$

Product rule to differentiate $x^2 e^{5x}$

M1

A1

$$\Rightarrow y'' = 2k e^{5x} + 10kx e^{5x} + 10kx e^{5x} + 25kx^2 e^{5x}$$

A1ft

$$\Rightarrow 2k e^{5x} + 20kx e^{5x} + 25kx^2 e^{5x} - 10(2kx e^{5x} + 5kx^2 e^{5x}) + 25kx^2 e^{5x} = 6e^{5x}$$

Substitution into differential equation

M1

A1

$$2k = 6 \Rightarrow k = 3$$

Only ft if $x e^{5x}$ and $x^2 e^{5x}$ terms all cancel out

A1ft

6

$$(b) \quad \text{Aux. eqn. } m^2 - 10m + 25 = 0 \Rightarrow m = 5$$

PI

B1

$$\text{CF is } (A + Bx)e^{5x}x$$

M1

Their CF + their / our PI

M1

$$\text{GS } y = (A + Bx)e^{5x}x + 3x^2 e^{5x}x$$

ft only on wrong value of k

A1ft

4

[10]

Q25.

IF is $e^{\int \tan x dx}$

M1

$$e^{-\ln \cos x} = e^{\ln \sec x}$$

Accept either

A1

$$= \sec x$$

ft on earlier sign error

A1ft

$$\frac{d}{dx}(y \sec x) = \sec^2 x$$

M1A1

$$y \sec x = \int \sec^2 x dx$$

$$y \sec x = \tan x + c$$

Condone missing c

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A1

$$y = 3 \text{ when } x = 0 \Rightarrow 3 \sec 0 = 0 + c$$

m1

$$c = 3 \Rightarrow y \sec x = \tan x + 3$$

OE; condone solution finishing at $c = 3$
provided no errors

A1

[8]

Q26.

(a) $(m + 1)^2 = -1$

Completing sq or formula

M1

$$m = -1 \pm i$$

A1

2

- (b) (i) CF is $e^{-x}(A \cos x + B \sin x)$
If m is real give M0

M1

{or $e^{-x}A \cos(x + B)$ **but not** $Ae^{(-1+i)x} + Be^{(-1-i)x}$ }

ft On wrong a 's and b 's but roots must be complex.

A1 ✓

{P.Int.} try $y = px + q$
OE

M1

$$2p + 2(px + q) = 4x$$

A1

$$p = 2, \quad q = -2$$

On one slip

A1 ✓

GS $y = e^{-x}(A \cos x + B \sin x) + 2x - 2$

Their CF + their PI with two arbitrary constants.

B1 ✓

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6

- (ii) $x = 0, y = 1 \Rightarrow A = 3$

Provided an M1 gained in (b)(i)

B1 ✓

$$y'(x) = -e^{-x}(A \cos x + B \sin x) +$$

Product rule used

M1

$$+ e^{-x}(-A \sin x + B \cos x) + 2$$

A1 ✓

$$y'(0) = 2 \Rightarrow 2 = -A + B + 2 \Rightarrow B = 3$$

Slips

A1 ✓

$$y = 3e^{-x}(\cos x + \sin x) + 2x - 2$$

4

[12]

Q27.

- (a) $y = x^3 - x \Rightarrow y'(x) = 3x^2 - 1$
Accept general cubic.

B1

$$\frac{dy}{dx} + \frac{2xy}{x^2 - 1} = 3x^2 - 1 + \frac{2x(x^3 - x)}{x^2 - 1}$$

Substitution into LHS of DE

M1

$$= 3x^2 - 1 + \frac{2x^2(x^2 - 1)}{x^2 - 1} = 5x^2 - 1$$

Completion. If using general cubic all unknown constants must be found

A1

3

- (b) $\frac{d}{dx}[(x^2 - 1)y] = 2xy + (x^2 - 1)\frac{dy}{dx}$

M1A1

Differentiating $(x^2 - 1)y = c$ wrt x

SC Differentiated but not implicitly give max of 1/3 for complete solution

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$$\text{leads to } 2xy + (x^2 - 1)\frac{dy}{dx} = 0$$

$$\Rightarrow y = \frac{c}{x^2 - 1} \text{ is a soln. of } \frac{dy}{dx} + \frac{2xy}{x^2 - 1} = 0$$

Be generous

A1

3

- (c) $\Rightarrow y = \frac{c}{x^2 - 1}$ is a soln with one arb. constant of $\frac{dy}{dx} + \frac{2xy}{x^2 - 1} = 0$

$$\Rightarrow y = \frac{c}{x^2 - 1} \text{ is a CF of the DE}$$

GS is CF + PI

Must be using 'hence'; CF and PI functions of x only

M1

$$y = \frac{c}{x^2 - 1} + x^3 - x$$

CSO

Must have explicitly considered the link between one arbitrary constant and the GS of a first order differential equation.

A1

2

[8]

Q28.

(a) (i) $y(1.1) = y(1) + 0.1[1 \ln 1 + 1/1]$

M1A1

$$= 1 + 0.1 = 1.1$$

A1

3

(ii) $y(1.2) = y(1) + 2(0.1)[f(1.1, y(1.1))]$

M1A1

$$\dots = 1 + 2(0.1)[1.1 \ln 1.1 + (1.1/1.1)]$$

On answer to (a)(i)

A1 ✓

$$\dots = 1 + 0.2 \times 1.104841198 \dots$$

$$\dots = 1.22096824 = 1.221 \text{ to 3 dp}$$

CAO

A1

4

(b) (i) IF is $e^{\int -\frac{1}{x} dx}$
Condone $e^{\int \frac{1}{x} dx}$ for M mark

M1

$$= e^{-\ln x}$$

A1

$$= e^{\ln x^{-1}} = x^{-1} = \frac{1}{x}$$

AG (be convinced)

(b)(i) Solutions using the printed answer must

A1

3

(ii) $\frac{d}{dx}\left(\frac{y}{x}\right) = \ln x$

M1A1

$$\frac{y}{x} = \int \ln x \, dx = x \ln x - \int x\left(\frac{1}{x}\right) dx$$

Integration by parts for $x^k \ln x$

M1

$$\frac{y}{x} = x \ln x - x + c$$

Condone missing c.

A1

$$y(1) = 1 \Rightarrow 1 = \ln 1 - 1 + c$$

Dependent on at least one of the two previous M marks

m1

$$\Rightarrow c = 2 \Rightarrow y = x^2 \ln x - x^2 + 2x$$

$$\text{OE eg } \frac{y}{x} = x \ln x - x + 2$$

A1

6

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(iii) $y(1.2) = 1.222543... = 1.223 \text{ to 3dp}$

B1

1

[17]

Q29.

(a) $y = 2x + \sin 2x \Rightarrow y' = 2 + 2 \cos 2x$
 $\Rightarrow y'' = -4 \sin 2x$

Need to attempt both y' and y''

M1 A1

$$-4 \sin 2x - 5(2 + 2 \cos 2x) + 4(2x + \sin 2x) = 8x - 10 - 10 \cos 2x$$

CSO AG Substitute. and confirm correct

A1

3

(b) Auxiliary equation $m^2 - 5m + 4 = 0$

M1

$$m = 4 \text{ and } 1$$

A1

$$\text{CF: } Ae^{4x} + Be^x$$

M1

$$\text{GS: } y = Ae^{4x} + Be^x + 2x + \sin 2x$$

Their CF + 2x + sin2x

B1ft

4

(c) $x = 0, y = 2 \Rightarrow 2 = A + B$
Only ft if exponentials in GS

B1ft

$$x = 0, y' = 0 \Rightarrow 0 = 4A + B + 4$$

*Only ft if exponentials in GS and
differentiated four terms at least*

B1ft

Solving the simultaneous equations

M1

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gives $A = -2$ and $B = 4$

A1

4

$$y = -2e^{4x} + 4e^x + 2x + \sin 2x$$

[11]

Q30.

(a) IF is $e^{\int \cot x dx}$

M1

$$= e^{\ln \sin x}$$

A1

$$= \sin x$$

AG

A1

3

(b) $\frac{d}{dx}(y \sin x) = 2 \sin x \cos x$

M1 A1

$$y \sin x = \int \sin 2x \, dx$$

Method to integrate $2 \sin x \cos x$

M1

$$y \sin x = -\frac{1}{2} \cos 2x + c$$

OE

A1

$$y = 2 \text{ when } x = \frac{\pi}{2} \Rightarrow 2 \sin \frac{\pi}{2} = -\frac{1}{2} \cos \pi + c$$

Depending on at least one M

m1

$$c = \frac{3}{2} \Rightarrow y \sin x = \frac{1}{2} (3 - \cos 2x)$$

OE eg $y \sin x = \sin^2 x + 1$

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A1

6

[9]

Q31.

(a) $u = \frac{dy}{dx} + 2y \Rightarrow \frac{du}{dx} = \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx}$

2 terms correct

M1

A1

$$\text{LHS of DE} \Rightarrow \frac{du}{dx} - 2 \frac{dy}{dx} + 4 \frac{dy}{dx} + 4y$$

$$\text{LHS: } \frac{du}{dx} + 2(u - 2y) + 4y$$

Substitution into LHS of DE as far as no derivatives of y

M1

$$\Rightarrow \frac{du}{dx} + 2u = e^{-2x}$$

CSO AG

A1

4

(b) IF is $e^{\int 2dx} = e^{2x}$

B1

$$\frac{d}{dx}[ue^{2x}] = 1$$

M1 A1

$$\Rightarrow ue^{2x} = x + A$$

A1

$$\Rightarrow u = xe^{-2x} + Ae^{-2x}$$

A1

5

Alternative : Those using CF+PI

Auxiliary equation, $m + 2 = 0 \Rightarrow u_{CF} = Ae^{-2x}$

B1

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For u_{PI} try $u_{PI} = kxe^{-2x} \Rightarrow$

M1

$$ke^{-2x} - 2kxe^{-2x} + 2kxe^{-2x} (= e^{-2x})$$

LHS

A1

$$\Rightarrow k = 1 \Rightarrow u_{PI} = xe^{-2x}$$

A1

$$\Rightarrow u_{GS} = Ae^{-2x} + xe^{-2x}$$

A1

(c) $\Rightarrow \frac{dy}{dx} + 2y = xe^{-2x} + Ae^{-2x}$

Use (b) to reach a 1st order DE in y and x

M1

IF is $e^{\int 2dx} = e^{2x}$

B1

$\Rightarrow \frac{d}{dx}[ye^{2x}] = x + A$

A1ft

$\Rightarrow ye^{2x} = \frac{x^2}{2} + Ax + B$

A1ft

$\Rightarrow y = e^{-2x} \left(\frac{x^2}{2} + Ax + B \right)$

A1

5

[14]

Q32.

Auxiliary eqⁿ. $m^2 + 121 = 0$

$m = \pm 11i$

Or knowing to quote S.H.M

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M1

C.F. $x = A \sin 11t + B \cos 11t$

A1

For PI $x = \alpha \sin \omega t + \beta \cos \omega t$

$\dot{x} = \alpha \omega \cos \omega t - \beta \omega \sin \omega t$

M1

$\ddot{x} = -\alpha \omega^2 \sin \omega t - \beta \omega^2 \cos \omega t$

M1

$-\alpha \omega^2 \sin \omega t - \beta \omega^2 \cos \omega t + 121 (\alpha \sin \omega t + \beta \cos \omega t) = 3 \sin \omega t$

$$-\alpha\omega^2 + 121\alpha = 3 \quad \Rightarrow \quad \alpha = \frac{3}{(121 - \omega^2)}$$

$$-\beta\omega^2 + 121\beta = 0 \quad \Rightarrow \quad \beta = 0$$

For α and β

A1F

$$x = A \sin 11t + B \cos 11t + \frac{3}{(121 - \omega^2)} \sin \omega t$$

B1F

[6]



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Examiner reports

Q2.

Most of the students were able to show that $\tan x$ was an integrating factor, although a significant minority failed to realise that $\int \frac{\sec^2 x}{\tan x} dx$ gave $\ln \tan x$ directly and instead wrote the integrand in the form $\cot x + \tan x$ and after integrating correctly applied the relevant log law and eventually, after further manipulation, showed that $\tan x$ was an integrating factor. The small number of students who wrote the integrand as $2\operatorname{cosec} 2x$ were generally less convincing in reaching the printed result. Almost all students showed that they knew how to use the printed integrating factor to start to solve the first order differential equation but a minority of these students could not then find a correct method to integrate $\tan^2 x$.

Q3.

This question, which tested the solution of a second order differential equation with the form of the particular integral given, was a good source of marks for many students; especially before the boundary conditions in part (c) became part of the test. The majority of students found the correct values for the constants a , b and c in part (a) but sign errors were sometimes seen; the most common being to equate $-4c$ to 8 instead of -8 from consideration of the coefficients of e^{-3x} . In part (b), most students showed that they knew how to find the complementary function and then the general solution. Part (c) caused students more problems than the previous two parts. Although most students scored the mark for correctly dealing with the given boundary condition at $x = 0$, a significant minority did not realise that the coefficient of e^x , in the expression for $\frac{dy}{dx}$, had to be 0 for the boundary condition $\frac{dy}{dx} \rightarrow -1$ as $x \rightarrow \infty$ to be satisfied. As mentioned in previous reports, the examiners expect students to give special attention to certain limits that are within the specification. In this question, the limit of xe^{-3x} as $x \rightarrow \infty$ needed to be treated in isolation, and this was not considered by a significant number of students.

Q4.

This question, which tested use of an integrating factor to solve a first-order differential equation, was a good source of marks for many students. The first two parts of the question were usually answered correctly and almost all students correctly reached the stage $y \ln x = \int 9x^2 \ln x dx$ in part (b)(ii). A large majority of students used integration by parts correctly but a small number forgot the '9' at this stage and so lost the two remaining accuracy marks. Very few failed to include the '+c' and almost all of those students who had scored the method mark for their integration also scored the method mark for correctly using the given boundary condition. Surprisingly there was a minority of students who failed to score the final mark due to a miscopy of an earlier correct expression when substituting their value for c , as illustrated by $y \ln x = 3x^2 \ln x - x^3 + 2e^3$.

Q5.

Most candidates were able to show that they knew how to find and use an integrating factor to solve a first order differential equation. The method of integration by parts was generally both spotted and applied correctly, although some failed to include the constant

of integration in their general solution, thus ending with a general solution of a first order differential equation which contained no arbitrary constant.

In part (b) the examiners expected candidates to pay particular attention to the limit of $x \ln x$ (or $x^3 \ln x$) as x tended to 0. Also some candidates were less than convincing in dealing with the term C / x^2 as x tended to 0. In general, part (b) was not answered as well as part (a).

Q6.

It was pleasing to see that once again most students were able to show that they knew how to find and use a correct integrating factor to solve a first order differential equation. The more common errors were failure to multiply the right-hand side by the integrating factor before integrating, forgetting to divide the constant of integration by the integrating factor when rearranging the general solution into the form $y = f(x)$ or even forgetting to include a constant of integration. Students who answered part (a) correctly were generally successful in part (b).

Q7.

Most candidates differentiated the given expression, substituted into the first order differential equation and equated coefficients to score the method marks. The most common error in solving the resulting equations was to write the solution of $26p = 13$ as $p = 2$.

In part (b), those candidates who wrote down the correct auxiliary equation, ' $m + 5 = 0$ ', generally

had no problems scoring all three marks. Those candidates who solved $\frac{dy}{dx} + 5y = 0$ sometimes forgot to include the constant of integration and so ended up with a general solution which had no arbitrary constant.

There was a minority of candidates who tried to use an integrating factor and appeared not to know the method of solving a first order differential equation by using complementary function and particular integral.

Q8.

Most candidates were able to find the correct values of the constants p and q so that the given expression was a particular integral of the differential equation. An even higher proportion of candidates scored full marks for their general solution in part (b) as examiners allowed follow through on incorrect values for p and q . However, less than 20% of the candidates scored full marks for their answers to part (c). This was almost entirely due to no explicit consideration being given to the value of xe^{-2x} as $x \rightarrow \infty$. More surprisingly, a significant minority of candidates could not correctly deal with the boundary

condition $\frac{dy}{dx} \rightarrow 0$ as $x \rightarrow \infty$ as they had an e^x term in their final answer.

Q9.

This question was answered well with most candidates able to show that they knew how to find and use a correct integrating factor to solve a first order differential equation. The majority of candidates then correctly used substitution or inspection to complete the integration. Those candidates who tried to integrate by parts often made errors or did not

appreciate that two applications were required. The minority of candidates who correctly expressed $\sin 2x \sin x$ as the difference of two cosines generally had less difficulty in completing the integration correctly.

Q10.

In part (a), it was pleasing to see a higher proportion of candidates than in previous papers using the given form of the particular integral rather than introducing the extra term qe^{-2x} . The majority of candidates used the product rule correctly to find the value of the constant p . Most candidates showed that they knew the methods required to solve the second order differential equation, although a minority of candidates made the error of applying the boundary conditions to the complementary function before adding on the particular integral.

Q11.

This unstructured question was answered well by many of the candidates. However, a significant minority of candidates did not even write down the correct form of the auxiliary equation, or they solved it incorrectly to give real rather than imaginary values. When finding the particular integral, a large number of candidates considered more terms than they needed to, and they did not always go on to show that the relevant coefficients of the extra terms were zero. The majority of candidates knew that they had to find a complementary function and a particular integral and almost all scored the final mark for combining the two to give the general solution.

Q12.

This was another very good source of marks for candidates, with many fully correct solutions presented. In part (a) some candidates decided to ignore the given form of the particular integral and worked with $p\cos 2x + q\sin 2x$. Such an approach was not penalised

by examiners provided the candidate showed that both $p = 0$ and $q = -\frac{1}{3}$. The vast majority of candidates showed that they knew the methods to solve the second order differential equation but errors in forming and solving the auxiliary equation were sources of loss of marks. Real solutions from the auxiliary equation were more heavily penalised than other errors.

Q13.

Approximately half the candidates scored full marks for this question. Almost all candidates were able to show that they knew how to find and use an integrating factor to solve a first order differential equation. However, a significant minority did not recognise

the form of the integral of $\int x^3 (x^4 + 3)^{\frac{3}{2}} dx$ as $k(x^4 + 3)^{\frac{5}{2}}$ or did not apply a suitable substitution, so made very little further progress.

Q14.

Most candidates were able to show that $\frac{1}{x^2}$ was an integrating factor for the given first-order differential equation and the majority of them went on to correctly find its general solution. A small minority of candidates failed to insert the constant of integration and ended with a general solution which contained no arbitrary constant.

Candidates are advised to check that their general solutions contain as many arbitrary constants as the order of the differential equation being solved. The error ' $y = x^{-2} (\ln x + c)$ '

following a correct previous line ' $\frac{y}{x^2} = \ln x + c$ ' was seen more often than expected.

Q15.

Most candidates were able to show that they knew how to find and use an integrating factor to solve a first order differential equation. However, a significant number failed to find the correct integrating factor because they missed the negative sign and used $e^{\int \tan x \, dx}$. Candidates should be aware that, unless told otherwise in the question, it is acceptable to leave the solution in a form other than $y = f(x)$. Some candidates lost the final accuracy mark because either they had attempted to divide throughout by $\cos x$ but forgot to divide the $+ c$ as well before substituting in the boundary condition, or had made the arithmetical

error " $2 = -\frac{1}{2} + c \Rightarrow c = \frac{3}{2}$ ".

Q16.

In part (a), many candidates decided to ignore the given form of the particular integral and worked with $a \cos x + b \sin x$. Such an approach was not penalised by examiners provided the candidate showed that both $a = 0$ and $b = 2$. The vast majority of candidates showed that they knew the methods required to solve the second order differential equation, but arithmetical errors in solving the auxiliary equation $m^2 + 2m + 5 = 0$ or in

applying the boundary condition $\frac{dy}{dx} = 4$ when $x = 0$ or the wrong differentiation of $\cos 2x$ were sources of loss of marks as well as the more serious error of applying the boundary conditions to the complementary function before adding on the particular integral.

Q17.

This question, which required candidates to solve a second order differential equation, was generally a good source of marks. It was disappointing to see some candidates trying to solve the auxiliary equation, $m^2 + 4m + 5 = 0$, by factorisation. They obtained real solutions and this error was penalised heavily. Better candidates were able to write down the correct complementary function and find the particular integral but some wasted valuable time by starting with $y_p = ax^2 + bx + c$ and showing that both a and b were zero. Candidates who were able to find the correct general solution in part (a) usually went on to apply the given boundary conditions correctly in their answers to part (b).

Q18.

Although many candidates were able to write down the integrating factor in terms of an

integral, a significant minority could not then integrate $\frac{4x}{x^2 + 1}$ correctly. Those who found the correct simplified integrating factor generally used it appropriately and either solved the resulting integral by a suitable substitution or, more frequently, just multiplied out and integrated $x^5 + 2x^3 + x$. Although some candidates failed to insert the constant of integration and so lost the final two marks, this was not a common error.

Q19.

Most candidates differentiated the given expression correctly and substituted the result into the given differential equation. Subsequently, however, there were many cases of an incorrect expansion for the term $-3y$ which lead to the values for two of the four constants being incorrect. In part (b), it was pleasing to find only a small number of candidates trying to use an integrating factor with the original differential equation. The successful candidates either used the auxiliary equation $m - 3 = 0$ or used the reduced equation and then separated the variables. A common error, after finding $m = 3$, was to assume that this represented repeated roots, leading to a complementary function of the form $e^{3x}(Ax + B)$, with two arbitrary constants for this first order differential equation. The error for those using the separation of variables method was to omit the arbitrary constant.

Q20.

Most candidates scored full marks for their solution to part (a). Again, part (b) was answered well by the majority of candidates, but it was not uncommon to find solutions which used the wrong integrating factor, x , or lacked an arbitrary constant. Those

candidates who started part (c) by equating their answer for (b) to $\frac{dy}{dx}$ and integrating normally scored both marks, although some lost the accuracy mark because their general solution of this second order differential equation did not contain two arbitrary constants.

Q21.

In part (a), many candidates scored high marks for finding the general solution of the second order differential equation, with the most common slip being a wrong expansion of brackets, which led to -9 instead of -3 for the constant part of the particular integral. A more serious but less common error was to look for a particular integral of the form $axe^{-2x} + b$. Part (b) proved, as expected, to be more of a challenge to candidates, with many unable to deal with the boundary condition expressed as a limit. This was the first time that this had appeared on an examination paper for this unit and, in general, only the better candidates could handle it.

Q22.

The vast majority of candidates answered part (a) correctly and also used it to write the left-hand side of the differential equation as an exact derivative. However, finding the

resulting integral, $\int 3x^2 (x^3 + 1)^{\frac{1}{2}} dx$, caused problems for a majority of candidates. It was disappointing to see so many failing to spot this integrand as the derivative of

$\frac{2}{3} (x^3 + 1)^{\frac{3}{2}}$ with some resorting to writing $(x^3 + 1)^{\frac{1}{2}}$ as $\left(x^{\frac{3}{2}} + 1\right)$. Subsequent credit, for method, was given to those candidates who introduced a constant of integration and found its value by applying the given boundary conditions.

Q23.

This question, on finding the general solution of a second order differential equation, was a good source of marks for many candidates. Some candidates failed to write their final answer in the form $y = f(x)$, but a more common error was not finding/stating "2" as part of the particular integral.

Q24.

Many candidates gained very high marks for this opening question which tested the solution of a second order differential equation. In part (a), although the form of the particular integral was given in the question, a significant minority of candidates preferred to discover it themselves and tried particular integrals $y = ae^{5x}$ and $y = axe^{5x}$, showing that each failed before trying $y = ax^2e^{5x}$ and finding that $a = 3$. These candidates were not penalised for this approach but clearly they used up valuable time in doing more than what was required. Some candidates also found the complementary function in part (a) and full relevant credit was given even if this work was not repeated in part (b). Most candidates indicated that they knew how to find the complementary function and how to use it to find the general solution but a significant number failed to score full marks because they did not give their final answer in the form ' $y = f(x)$ ' where $f(x)$ contains two arbitrary constants.

Q25.

This question on solving a first-order differential equation by use of an integrating factor was the best-answered question on the paper. Only a few candidates failed to find a correct integrating factor; such candidates failed to realise that the integral of $\tan x$ is available from the AQA formulae booklet. The most common errors were forgetting to multiply the right-hand side of the given differential equation by $\sec x$, or rearranging ' $y \sec x = \tan x + c$ ' incorrectly as ' $y = \tan x \cos x + c$ ' before substituting the given boundary conditions.

Q26.

The majority of candidates gained at least ten of the twelve available marks for this opening question which tested the solution of a second order differential equation. Part (a) caused few problems although there were a few who had difficulty simplifying

$$\frac{\sqrt{4-8}}{2}$$

correctly. The most common method error in the question occurred in part(b)(i). These candidates considered a particular integral of the form ' ax ' instead of the correct form ' $ax + b$ '. In general, candidates were able to apply the given boundary conditions in a correct manner.

Q27.

A significant number of candidates attempted to solve the differential equation by using an integrating factor. Some were clearly mixing up 'integrating factor' with 'particular

integral' as their solutions stated 'particular integral = $e^{\int \frac{2x}{x^2-1} dx}$ ', whereas others seemed to almost go onto 'auto-pilot' as they just solved the differential equation without reference to a particular integral or implicit differentiation. For full marks the examiners expected candidates to justify that the solution given in part (b) was indeed the complementary function. A statement linking one arbitrary constant with the general solution of a first order differential equation was required.

Q28.

This question was a good source of marks for many candidates. The methods in part (a) were well understood and most candidates worked accurately to obtain the correct approximations. In part (b)(i) a high proportion of the candidates obtained the printed answer but some failed to show sufficient stages in their working. In part (b)(ii), weaker

candidates could not integrate $\ln x$; such candidates frequently gave the answer as $1/x$. Those who inserted the constant of integration were generally able to apply the given boundary condition. Surprisingly, not all candidates who obtained the correct solution for part (b)(ii) were able to give the correct value for y in part (b)(iii).

Q29.

Most candidates gained very high marks for this opening question which tested the solution of a second order differential equation. In part (a) it was acceptable to differentiate the given result and substitute into the given differential equation. Although sign errors were seen, the vast majority who used this approach scored all three marks. Those who took a general form for the particular integral sometimes failed to find all the unknown constants.

Part (b) was generally answered correctly although some gave the final answer as $y = Ae^{4x} + Be^{2x} + 8x - 10 - 10\cos 2x$. Except for some careless slips in using $y'(0) = 0$ the final part of the question was answered well.

Q30.

The vast majority of candidates started correctly by writing $= e^{\int \cot x \, dx}$, and most went on to score the 3 marks. Some candidates, however, seemed to be unaware that $\int \cot x \, dx$ is given in the formulae booklet and can be quoted directly. In part (b) some candidates had problems integrating $2\sin x \cos x$, whilst others used inspection, converting to $\sin 2x$, substitution or integration by parts. Those who then divided throughout by $\sin x$ to get y explicitly sometimes failed to divide the term '+ c'. In finding the constant of integration it was not uncommon to see $-\frac{1}{2} \cos \pi$ evaluated incorrectly as $-\frac{1}{2}$ or even 0.

Q31.

This was a good source of marks for many candidates. Part (a) was answered well although lengths of solutions varied greatly and a few candidates failed to insert sufficient steps to convince the examiners that the printed answer had been obtained convincingly. In part (b), although the complementary function/particular integral approach was applied correctly by a number of candidates, use of the integrating factor was generally more successful. However some weaker candidates failed to multiply the right-hand side by the integrating factor before proceeding. The start of part (c) begins with 'Hence' without an 'or otherwise' so candidates were expected to use the previous results. Those candidates who tried to solve the differential equation directly generally failed to find the correct particular integral. Many candidates who used the correct method went on to score full marks but some others forgot to insert the constant of integration and ended with the general solution of a second-order differential equation with only one arbitrary constant.

Also, surprisingly, $\int^x dx$, was not always found correctly.